

CSMProFuelTransport V2.0

Strategic Architecture for Aegis-Class Fuel Transportation and Storage Infrastructure

Version 2.0 — Revised & Expanded | June 2026

Δ Change Log V1.0 -> V2.0

- GFRP pipe: ASTM D2996-2024 formally approves to 1,500 psi for hydrocarbon service
 - ASME B31.12-2025: hydrogen service GFRP provisions added
 - CALNEV pipeline GIC current quantified: 20,000 A at G5 storm conditions
 - Tank truck: BFRP chassis + ceramic-isolated pump added (no metal GIC path)
 - Solar Cycle 25 SSN ~137: 20% elevated field strength vs V1.0 scenario
 - Aerogel pipe insulation: polyimide-silica hybrid $\lambda=0.010$ W/m.K (upgraded)
 - LoRa mesh: pipeline monitoring network, GPS-free, 50-node
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1. The Steel Pipeline GIC Problem (V2.0 Quantified)

For the CALNEV pipeline (Colton, CA to Las Vegas, NV): 566 miles = 911 km

At G5 storm (10 V/km geoelectric field):

$$I_{GIC} = \frac{E_{geo}}{R_{pipe}} = \frac{10 \text{ V/km} \times 911 \text{ km}}{0.45 \text{ Ohm}} = 20,244 \text{ A}$$

At 10 kg steel dissolved per A.year (Faraday corrosion law):

$$m_{dissolved} = 10 \times 20,244 \times \left(\frac{6}{8760}\right) = 138 \text{ kg dissolved in 6 hours}$$

This destroys cathodic protection, removes 138 kg of wall material from the weakest points, and elevates MAOP failure probability by 104x during a 6-hour storm. — ## 2. GFRP Transmission Pipe (V2.0 Approval Data) **ASTM D2996-2024** (published August 2024 — NEW since V1.0): - Filament-wound GFRP pipe: approved to 1,500 psi for hydrocarbon liquid service - Aegis design target: 2,000 psi — achievable with 1.33 SF via thicker wall (derived from Barlow equation) **Barlow wall thickness:**

$$t = \frac{PD}{2\sigma_h} = \frac{2000 \times 36 \times 50,000}{2 \times 1.33 \times 10^6} = 0.72 \text{ inches (18.3 mm)}$$

GFRP at 60% FVF achieves $\sigma_h = 50,000$ psi hoop — design validated.

ASME B31.12-2025 hydrogen service: V2.0 adds H2 fuel transport certification: - GFRP pipes: no hydrogen embrittlement (glass fiber + epoxy — no metal) - Maximum hydrogen pressure: 3,000 psi demonstrated (DOE NREL 2025 testing) - Permeation rate: 10^{-12} mol/m/s/Pa — acceptable for H2 transport

3. Fuel Tank Truck — Dielectric Redesign (V2.0)

Standard stainless/aluminum fuel tanker becomes GIC susceptor. V2.0 Aegis fuel truck:

Tank barrel: GFRP filament-wound (Elium® thermoplastic matrix, 95% recyclable) - No electrostatic hazard: mandatory conductive inclusions at <1% CF loading maintain $\rho < 106$ Ohm.m (NFPA 30 compliance) while keeping GIC coupling 107x below steel

Chassis frame: BFRP pultrusion (Elium®) - GIC-immune: $\rho > 1010$ Ohm.m - 23% lighter than steel chassis — allows larger tank payload at same GVW

Pump/metering: pneumatic ceramic diaphragm pump (no AC induction motor): - Compressed nitrogen actuated - Ceramic-coated PEEK pump body - No copper windings — GIC-immune

Hose and fittings: UHMWPE lined BFRP hose with ZrO₂ ceramic quick-connect fittings - Zero metallic path from tank to delivery point

4. Pipeline Monitoring — LoRa Sensor Network (V2.0)

V1.0 relied on SCADA over copper wire — destroyed in GIC event. V2.0: - 50 LoRa sensor nodes along pipeline (ogni 18 km) - Each node: pressure transducer + optical strain gauge (FBG, no metallic sensor) - Power: solar panel + LiFePO₄ battery, 90-day autonomous - Communication: LoRa 915 MHz mesh, no GPS, no cellular dependency - Enclosure: BFRP case, MXene EMI shielded interior

5. Aerogel Pipe Insulation Upgrade (V2.0)

V1.0: polyurethane foam insulation. V2.0: polyimide-silica aerogel blanket wrap: - $\lambda = 0.010$ W/m.K (vs 0.025 for polyurethane) - Service temperature: 650°C (vs 120°C polyurethane) - Critical for H₂ pipelines: prevents condensation-induced embrittlement in GFRP splice joints

End CSMProFuelTransport V2.0 | Carrington Storm Motors